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Unit 4 : Session 10 : SLO 2 | Client-Server Communication Using the socket Module

I. Steps to Create a Simple TCP Server

1. Create the socket: `s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)` # IPv4 + TCP
2. Bind it to an address: `s.bind(("localhost", 5000))`
3. Start listening: `s.listen(1)` # backlog of 1 pending connection
4. Accept a client: `conn, addr = s.accept()` # blocks until a client connects
5. Exchange data: `data = conn.recv(1024); conn.send(b"Hello client")`
6. Close both sockets: `conn.close(); s.close()`

II. Minimal Working Example

- `import socket`
- `s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)`
- `s.bind(("localhost", 5000))`
- `s.listen(1)`
- `conn, addr = s.accept()`
- `print("Connected:", addr)`
- `conn.sendall(b"Welcome")`
- `conn.close(); s.close()`